

Questions

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Q1 - 2024 (04 Apr Shift 1)

The equation of stationary wave is :

$$y = 2a \sin\left(\frac{2\pi nt}{\lambda}\right) \cos\left(\frac{2\pi x}{\lambda}\right).$$

Which of the following is NOT correct :

(1) The dimensions of n/λ is [T]

(2) The dimensions of n is $[LT^{-1}]$

(3) The dimensions of x is [L]

(4) The dimensions of nt is [L]

Q2 - 2024 (04 Apr Shift 2)

Applying the principle of homogeneity of dimensions, determine which one is correct,

where T is time period, G is gravitational constant, M is mass, r is radius of orbit.

(1) $T^2 = \frac{4\pi^2 r^2}{GM}$

(2) $T^2 = \frac{4\pi^2 r}{GM^2}$

(3) $T^2 = \frac{4\pi^2 r^3}{GM}$

(4) $T^2 = 4\pi^2 r^3$

Q3 - 2024 (05 Apr Shift 1)

If G be the gravitational constant and u be the energy density then which of the following quantity have the dimensions as that of the $\sqrt{uG}$:

(1) pressure gradient per unit mass

(2) Gravitational potential

(3) Energy per unit mass

(4) Force per unit mass

Q4 - 2024 (05 Apr Shift 2)

Questions

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What is the dimensional formula of ab^{-1} in the equation $\left(P + \frac{a}{v^2}\right)(V - b) = RT$, where letters have their usual meaning.

(1) $[M^{-1} L^5 T^3]$

(2) $[M^6 L^7 T^4]$

(3) $[ML^2 T^{-2}]$

(4) $[M^0 L^3 T^{-2}]$

Q5 - 2024 (06 Apr Shift 1)

Match List I with List II

LIST I	LIST II		
A.	Torque	I.	$[M^1 L^1 T^{-2} A^{-2}]$
B.	Magnetic field	II.	$[L^2 A^1]$
C.	Magnetic moment	III.	$[M^1 T^{-2} A^{-1}]$
D.	Permeability of free space	IV.	$[M^1 L^2 T^{-2}]$

Choose the correct answer from the options given below:

(1) A-III, B-I, C-II, D-IV

(2) A-IV, B-II, C-III, D-I

(3) A-IV, B-III, C-II, D-I

(4) A-I, B-III, C-II, D-IV

Q6 - 2024 (06 Apr Shift 2)

Given below are two statements :

Statement (I) : Dimensions of specific heat is $[L^2 T^{-2} K^{-1}]$.Statement (II) : Dimensions of gas constant is $[ML^2 T^{-1} K^{-1}]$.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both statement (I) and statement (II) are correct

(2) Statement (I) is correct but statement (II) is incorrect

(3) Both statement (I) and statement (II) are incorrect Statement (I) is incorrect but statement (II) is correct

Questions

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(4) Statement (I) is incorrect but statement (II) is correct

Q7 - 2024 (08 Apr Shift 2)

If ϵ_0 is the permittivity of free space and E is the electric field, then $\epsilon_0 E^2$ has the dimensions :

(1) $[M^{-1} L^{-3} T^4 A^2]$

(2) $[ML^2 T^{-2}]$

(3) $[M^0 L^{-2} T A]$

(4) $[ML^{-1} T^{-2}]$

Q8 - 2024 (09 Apr Shift 1)

The dimensional formula of latent heat is :

(1) $[ML^2 T^{-2}]$

(2) $[M^0 L^2 T^{-2}]$

(3) $[MLT^{-2}]$

(4) $[M^0 LT^{-2}]$

Q9 - 2024 (09 Apr Shift 2)

The de-Broglie wavelength associated with a particle of mass m and energy E is $h/\sqrt{2mE}$. The dimensional formula for Planck's constant is :

(1) $[ML^2 T^{-1}]$

(2) $[ML^{-1} T^{-2}]$

(3) $[MLT^{-2}]$

(4) $[M^2 L^2 T^{-2}]$

Questions

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Answer Key

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Q1 (1) mathongo Q2 (3) mathongo Q3 (4) mathongo Q4 (3) mathongo

Q5 (3) mathongo Q6 (2) mathongo Q7 (4) mathongo Q8 (2) mathongo

Q9 (1) mathongo mathongo mathongo mathongo mathongo mathongo

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Solutions

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Q1

Comparing the given equation with standard equation of standing $\frac{2\pi n}{\lambda} = \omega$ & $\frac{2\pi}{\lambda} = k$

$$\left[\frac{n}{\lambda}\right] = [\omega] = T^{-1}$$

$$[nt] = [\lambda] = L$$

$$[n] = [\lambda\omega] = LT^{-1}$$

$$[x] = [\lambda] = L$$

Q2

According to principle of homogeneity dimension of LHS should be equal to dimensions of RHS so option (3)

is correct.

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

$$[T^2] = \frac{[L^3]}{[M^{-1} L^3 T^{-2}] [M]}$$

(Dimension of G is $[M^{-1} L^3 T^{-2}]$)

$$[T^2] = \frac{[L^3]}{[L^3 T^{-2}]} = [T^2]$$

Q3

$$[uG] = [(M^1 L^{-1} T^{-2}) (M^{-1} L^3 T^{-2})]$$

$$[uG] = [M^0 L^2 T^{-4}]$$

$$[\sqrt{uG}] = [L^1 T^{-2}]$$

Q4

$$\therefore [V] = [b]$$

$$\therefore \text{Dimension of } b = [L^3]$$

$$[P] = \left[\frac{a}{V^2}\right]$$

$$[a] = [PV^2] = [ML^{-1} T^{-2}] [L^6]$$

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Solutions

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$$\text{Dimension of } a = [ML^5 T^{-2}]$$

$$\therefore ab^{-1} = \frac{[ML^5 T^{-2}]}{[L^3]} = [ML^2 T^{-2}]$$

Q5

$$[\vec{\tau}] = [\vec{r} \times \vec{F}] = [ML^2 T^{-2}]$$

$$[F] = [qVB]$$

$$\Rightarrow B = \left(\frac{F}{qV} \right) = \left[\frac{MLT^{-2}}{ATLT^{-1}} \right] = [MA^{-1} T^{-2}]$$

$$[M] = [I \times A] = [AL^2]$$

$$B = \frac{\mu_0 I dl \sin \theta}{4\pi r^2}$$

$$\Rightarrow [\mu] = \left[\frac{Br^2}{Idl} \right] = \left[\frac{MT^{-2} A^{-1} \times L^2}{AL} \right]$$

$$= [MLT^{-2} A^{-2}]$$

Q6

$$\Delta Q = mS\Delta T$$

$$s = \frac{\Delta Q}{m\Delta T}$$

$$[s] = \left[\frac{ML^2 T^{-2}}{MK} \right]$$

$$[s] = [L^2 T^{-2} K^{-1}]$$

Statement-(I) is correct

$$PV = nRT \Rightarrow R = \frac{PV}{nT}$$

$$[R] = \frac{[ML^{-1} T^{-2}][L^3]}{[mol][K]}$$

$$[R] = [ML^2 T^{-2} mol^{-1} K^{-1}]$$

Statement-II is incorrect

Q7

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Solutions

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$$E = \frac{KQ}{R^2}$$

$$E = \frac{Q}{4\pi\epsilon_0 R^2}$$

$$\epsilon_0 = \frac{Q}{4\pi R^2 E}$$

$$\text{Now, } \epsilon_0 E^2 = \frac{Q}{4\pi R^2 E} \cdot E^2 = \frac{Q}{4\pi R^2} \cdot E$$

$$[\epsilon_0 E^2] = \left[\frac{QE}{R^2} \right] = \frac{[Q][E]}{[R^2]} = \frac{[Q]}{[R^2]} \frac{[W]}{[Q][R]}$$

$$= \frac{[W]}{[R^3]} = \frac{ML^2 T^{-2}}{L^3} = ML^{-1} T^{-2}$$

Q8

Latent heat is specific heat

$$\Rightarrow \frac{ML^2 T^{-2}}{M} = M^0 L^2 T^{-2}$$

Q9

$$\lambda = \frac{h}{\sqrt{2mE}} \text{ or } E = h\nu$$

$$[ML^2 T^{-2}] = h [T^{-1}]$$

$$h = [ML^2 T^{-1}]$$

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